

ZAIN ALI BUTT

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FULL-STACK SOFTWARE DEVELOPER

Full-stack developer who takes client work from requirements through deployment and ongoing support. Delivered paid systems in 2026, with ongoing support. Strongest in TypeScript, Node and PostgreSQL, and in the integration and deployment work around them.

COMMERCIAL EXPERIENCE

Freelance Software Developer

Jun 2026 – Present

Self-Employed

London, UK

- Shipped a paid shipment-operations platform for a cross-border logistics team on Next.js, TypeScript, Supabase and Vercel. Retained for ongoing support.
- Replaced manual tracking and chasing with a single completion workflow. Client reported roughly 60% higher operational efficiency and near-elimination of routine tracking errors.
- Built and launched a booking-enabled website for a private music tutor. Client reported two-to-threefold growth in engagement, enquiries and bookings.

Invigilator

Nov 2024 – Present (seasonal)

Loxford School

London, UK

- Run regulated examinations for 10–220 candidates, covering access arrangements, incidents and secure materials under strict confidentiality.

PROJECTS

ProofLens | *Media provenance and tamper evidence*

Oct 2025 – Apr 2026

github.com/zainalibutt/Proof-Lens · proof-lens.vercel.app

- Built and deployed a provenance system across a React Native app, a 35-endpoint Express API, PostgreSQL/Supabase, AWS S3 and a public web verifier.
- Implemented SHA-256 hashing, device-held Ed25519 signing, independent server re-hashing and RFC 3161 timestamping, verifiable offline with OpenSSL.
- Research prototype, evaluated over 36 checks and seven investigations: 124 credentials anchored, 9.6-second mean capture-to-anchor.

River | *Multiplayer poker with solver-backed bots*

Aug 2026 – Present

playriver.uk

- Built and launched an early v1 of a server-authoritative Texas Hold'em platform: a pure TypeScript rules engine, a WebSocket server that owns every chip through an append-only ledger, and commit-reveal shuffles. 909 tests in CI.
- Took the table bots from rule-based play to a river solver with opponent modelling, measured against a worst-case exploiter and the public Slumbot benchmark (+24.5 bb/100 over the previous version).

Melody | *Company research terminal*

2026

github.com/zainalibutt/Melody · melody-terminal.vercel.app

- Built a command-first terminal over SEC EDGAR and delayed market data in React and Express, surfacing source, freshness and provider-failure states explicitly.

Lancaster Music Hall | *Venue booking and scheduling system*

Jan – Apr 2025

- Led six developers through requirements, delivery recovery, testing and handover, rebuilding the calendar, conflict-validation and analytics features in Java and MySQL.

EDUCATION

City St George's, University of London

2023 – 2026

BSc Computer Science — Upper Second-Class Honours (69.3%)

London, UK

Final-year project 83%. Selected modules: Theory of Computation 85.8%, Object-Oriented Analysis & Design 84.4%, Algorithms 83%, C++ 82.5%.

TECHNICAL SKILLS

Applied: TypeScript, JavaScript, React, Next.js, React Native, Node.js/Express, WebSockets, PostgreSQL/Supabase, REST APIs, AWS S3, GitHub Actions, Playwright, Vercel, Railway, Blender (Python).

Foundational: Python, Java, C++, machine learning, game-theoretic solvers, data structures and algorithms, databases, networking, operating systems and system architecture.